# Lily Oglesby

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## Purpose

Graduate student in business research with a background in data science and astrophysics. Interested in using data to solve real-world problems and support better decisions across industries. Experience building Python and SQL workflows, analyzing complex datasets, automating data processes, and communicating findings.

## Skills

- **Programming and Data Analysis:** Python, SQL, Pandas, NumPy, SciPy, Scikit-Learn, TensorFlow, Pytorch, Matplotlib, Seaborn, Jupyter
- **Databases and Development:** PostgreSQL, MongoDB, Git, GitHub, Linux, Docker, VS Code, JavaScript, HTML, CSS, LaTeX
- **Analytics and Modeling:** Data Cleaning and Validation, Exploratory Data Analysis, Statistical Modeling, Regression, Time-Series Analysis, Trend Analysis, Data Visualization, Machine Learning, Causal Inference, Bayesian Inference, Feature Engineering, Model Evaluation
- **Communication and Collaboration:** Technical and nontechnical communication, research communication, stakeholder communication, teaching and mentoring, independent and collaborative project work

## Education

- **University of Southern California | Marshall School of Business** — Los Angeles, CA
  *Master of Science in Business Research* — *Aug 2025 – Dec 2026*
  - **Fall 2026 coursework:** Business Analytics, AI Modeling for Decision-Making, Regression Analysis, Spreadsheet Modeling, Data Analytics; Tableau and SAS coursework in progress.
- **University of California, Berkeley** — Berkeley, CA
  *Bachelor of Arts: Double Major in Data Science and Astrophysics* — *Aug 2021 – May 2025*

## Selected Data and Analytics Projects

- **Interactive Portfolio Game**
  *JavaScript, HTML/CSS, Canvas, GSAP, Howler.js, Tiled* — *2026*
  - Built and debugged a responsive browser-based portfolio game with an explorable map, collision detection, turn-based battles, mobile controls, animation, audio, asset preloading, and production deployment.
- **Haqdarshak Social Protection Access Consulting Project**
  *USC Graduate Research, Data Analysis, Program Design, Stakeholder Analysis* — *2025*
  - Reviewed program and operational data to examine field-worker activity and barriers to connecting eligible individuals with government benefit programs; identified patterns and developed recommendations for improving outreach and service delivery.
- **Astronomy Data Analytics Portfolio**
  *Python, Pandas, NumPy, Scikit-Learn, PyMC, Data APIs, Astropy* — *2024*
  - Built reproducible workflows for satellite observations, stellar time-series data, telescope images, and astronomical spectra, using data queries, preprocessing, regression, cross-validation, Bayesian estimation, and statistical visualization.
- **Particulate Matter Health Impact and Forecasting**
  *Python, Pandas, Scikit-Learn, Statistical Modeling, Causal Inference* — *2024*
  - Analyzed CDC and NOAA datasets to study relationships between pollution exposure and asthma outcomes and predict pollution from weather conditions; compared statistical and machine-learning models using leakage-resistant temporal validation.
- **PostgreSQL vs. MongoDB Yelp Benchmark**
  *PostgreSQL, MongoDB, Python, Jupyter, Docker* — *2024*
  - Designed equivalent analytical workloads across relational and document databases and compared data models, SQL queries, aggregation pipelines, indexing strategies, query plans, scan volume, and runtime performance.

## Research Experience

- **Space Sciences Laboratory, UC Berkeley** — Berkeley, CA
  *Undergraduate Researcher – Data Analysis* — *Aug 2021 – May 2025*
  - Built reusable Python workflows for three years of satellite data, applying time-series preprocessing, statistical filtering, validation, and visualization to identify event-related patterns and support a peer-reviewed publication in the *Journal of Geophysical Research*.
- **Institute for Astronomy, University of Hawai'i at Manoa** — Honolulu, HI
  *Data Science and Machine Learning Research Intern* — *Jun 2024 – Aug 2024*
  - Analyzed approximately 120,000 astronomical images using anomaly-detection workflows, trained a convolutional neural network for multiclass classification, and used GPU acceleration to improve preprocessing and model execution efficiency. Report available online.

## Teaching and Mentoring Experience

- **USC, UC Berkeley, and University of Hawai'i** — Los Angeles, Berkeley, and Honolulu
  *Instructor, Teaching Assistant, Reader, Grader, and Research Mentor* — *Aug 2022 – Jul 2026*
  - Taught and mentored high school, undergraduate, and graduate students through USC, UC Berkeley, and the Mauna Kea Scholars Program; led workshops and office hours on Python, data analysis, responsible AI, research design, graduate applications, and technical presentations, adapting instruction to different backgrounds and experience levels.